

Hospital Data Warehouse Engineering: An ETL Pipeline Implementation Using Pentaho and Star Schema Design

Marcellino Wilson Rusli
School Of Computer Science
BINUS University
Jakarta, Indonesia
marcellino.rusli@binus.ac.id

Jason Alexander Wijaya
School Of Computer Science
BINUS University
Jakarta, Indonesia
jason.wijaya013@binus.ac.id

Timothy Jason Latief
School Of Computer Science
BINUS University
Jakarta, Indonesia
timothy.latief@binus.ac.id

Abstract — In this paper, the design and implementation of the data engineering pipeline for the Indonesian national hospital registry is discussed. The scope includes the Extract, Transform, Load (ETL) process implemented with Pentaho Data Integration, the creation of a star schema data warehouse deployed in MySQL, and Power BI as an interactive analytics consumption layer. The source dataset consists of 3,155 rows of information about hospitals across 38 provinces of Indonesia, including types of facilities, classes, ownership and BLU/BLUD finance status, and capacity measurements (beds, services, staff). After cleaning, 3,147 rows (99.7%) were imported into the data warehouse; the eight rows rejected were due to sourcedata issues that could be easily identified. The next phase of referential integrity check revealed two distinct data quality issues: the presence of an unkeyed upsert procedure in the location dimension and dimensiongrain misalignment in the ownership dimension, both of which are mentioned explicitly and serve as an evaluation point in this paper. Power BI provides users with access to eleven reports and seven DAX measures based on the warehouse, showing significant geographic unevenness of the healthcare workforce and beds available with Java having the largest share and several Papua provinces showing the smallest numbers nationally.

Keywords — Data Engineering, ETL Pipeline, Data Warehouse, Star Schema, Pentaho, Power BI, Data Quality

I. INTRODUCTION

The growing volume of health information in Indonesia calls for a need for an efficient method for data engineering whereby operational data will be transformed into business intelligence. For example, while undertaking research on national health policies concerning Indonesia's National Health Insurance (JKN), it has become apparent that ratios of facilities and specialist physicians are very biased towards Java/Bali as compared to the other regions, and there is also a difference of factors of ten between the supply side variables in the various districts.[8]

In the context of the current project, the Hospital Management System has produced a massive amount of data, which is heterogeneous and covers many administrative levels

such as facility types, geographic locations, ownerships, capacity, and manpower data.

The project attempts to solve the problem of development of an end-to-end data warehouse for the national hospital register of Indonesia, which is a database with 3,155 entries about hospitals in 38 Indonesian provinces. The primary goal is to create a complete data engineering pipeline that will allow analysis of the healthcare infrastructure of Indonesia in terms of its capacity and personnel distribution among regions.

There are three contributions of this project to the research area. First, the project describes in a reproducible way a pipeline for the nationally representative dataset, with every transformation step explained in details. Secondly, instead of trusting in the design of the pipeline, the output of the pipeline is verified using an independent referential integrity audit and real data quality problems are revealed. Thirdly, Power BI consumption layer is discussed in terms of measures and report pages. All together, these three parts form a more comprehensive and well-evaluated pipeline than usually seen in applied academic research.

II. LITERATURE REVIEW

A. Data Engineering Architecture

The architecture of data engineering is an overall design and structure of processes involved in data collection, storage, processing, and consumption in enterprise applications. A typical architecture of a pipeline includes the following five components: (1) sources of raw data; (2) data ingestion, which is the process of extracting data from sources into processing systems; (3) data storage, which may come in various forms such as a data warehouse for analytics or a data lake for storing data in different formats; (4) data transformation, which is the

process of converting raw data into analytical data; and (5) data consumption.

B. ETL vs ELT

ETL and ELT (Extract, Load, and Transform) are the two prevailing paradigms used in data integration. In ETL, transformation takes place before loading data in the target storage using a specialized transformation engine. This paradigm is appropriate for structured data of small-medium volume, provides better data privacy control, but is slower and not as flexible. ELT includes loading of raw data into the target store before transformation using the computational capabilities of the warehouse; this paradigm is appropriate for large-scale cloud-native environment and highly flexible as far as re-transformation is concerned. ETL tools include Talend, SSIS, Informatica, and Pentaho. The tools for ELT include dbt, BigQuery, Snowflake, Databricks, and large-scale batch engines from the Hadoop ecosystem for high-volume and less structured sources [5][3].

In the current research, ETL was chosen due to structured nature of the hospital dataset, its medium volume (3,155 records), and necessity to cleanse and standardize data before loading in the warehouse.

C. Slowly Changing Dimensions

Dimension tables in data warehouses can evolve over time. In Slowly Changing Dimension Type 1 (SCD1), updates overwrite existing records, making history unavailable; this technique is suitable for cases where history is not significant analytically. On the other hand, SCD Type 2 inserts records to ensure that history is available, albeit at the cost of additional storage. The Insert/Update operation used in all the dimension transformation exercises done in this project is SCD1 if the natural key is set up correctly.

D. Related Work on Hospital Data Engineering in Indonesia

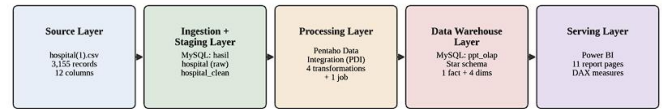
Several Indonesian papers have helped inform the decisions in this paper. Mariana et al. calculated Bed Occupancy Rate, Average Length of Stay, Turn Over Interval, and Bed Turn Over from the daily number of inpatients in seven wards and found that capacity management can be better viewed with the help of dynamic capacity measures rather than simply the static measure of beds [9]; this represents an additional approach to the static capacity measures (beds, services, manpower) used in the current warehouse. Purwanto and Renny have created a hospital data warehouse using Kimball's ninestep approach along with a fact constellation structure consisting of three fact tables and eleven dimensions based on SIMRS data [10].

The current study, on the other hand, uses the singlefact star schema due to the more limited nature of analysis of a national facility registry compared to encounter level transactional data. In a study by Prasetya and Kurniawan, an ETL process was designed for a retail sales data warehouse, which used the

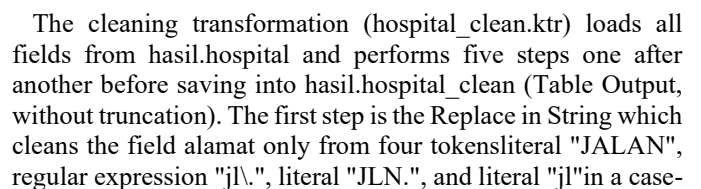
stepwise approach of filter, deduplicate, and look-up-based key matching utilized in the current study, but in a different context [11]. Most relevantly, Nurhakim and Voutama developed a Power BI dashboard from an ETLprocessed dataset in the health care industry in order to analyze service efficiency, which concluded that interactive slicers, KPI cards, and bar charts were efficient in communicating information from healthcare datasets [12].

III. METHODOLOGY

A. Flow Chart



The end-to-end pipeline is organized into five sequential layers. The Source Layer contains the raw hospital.csv file as the single point of data ingestion. The Ingestion and Staging Layer reads that file into a MySQL staging table via a Pentaho transformation and is designed to be fully idempotent through the use of a truncate-before-load strategy. The Processing Layer applies a series of ordered cleansing operations in a second Pentaho transformation, producing a curated staging table that serves as the input to the warehouse load. The Data Warehouse Layer is a star-schema model hosted in a separate MySQL schema, populated by four independent dimension transformations followed by a single fact-load transformation, all coordinated by one Pentaho job. Finally, the Serving Layer is Power BI Desktop, which connects directly to the warehouse schema and delivers interactive dashboards to end-users. Each layer has a clearly bounded interface to the next, and every stage is designed to be independently re-runnable without propagating errors downstream.



insensitive way. In the second step, the String Operations changes the fieldspropinsi, kab, and alamat to uppercase, the fieldjenis to lowercase, and the fieldkelas to uppercase. At that step, the fieldsnama, status_blu, and kepemilikan remain unchanged. In the third and fourth steps, two Filter Rows transform filtersrows based on conditionsstatus_blu like "BLU" and jenis like "rumah sakit". Afterwards, rows are filtered again by kelas being equal to either A, B, C, D, or D Pratama. Finally, a step Unique Rows (HashSet) excludes duplicate rows before Table Output.

The exclusion of the eight rows in the 3,155-row raw data set is due to the problems associated with the sourcedata itself. Certain rows contained a status_blu attribute whose value was either null or empty meaning that there had been no BLU or BLUD classification for financial management in the source system for that particular facility. Other rows contained a jenis attribute that indicated a non-hospital facilities like clinic or community health center which were incorrectly extracted into the hospital registry extract. Another category had a kelas attribute whose value was out of the five recognized hospital classifications in the source system probably as legacy codes or erroneous data entry. The one row of exclusion was a full row duplicate of another row due to double entry of data and was therefore deleted in the unique rows step.

E. Data Quality Validation

Apart from the row count check which is already part of the cleansing transformation in itself, the built warehouse has been independently verified for referential integrity as well. This has been done by recreating every match condition for each Pentaho Database Lookup against the database snapshot provided along with this report: for each fact row, the natural keys that it holds have been cross tabulated against the natural keys held by the dimension tables for verifying if there will be any resolution to these dimensional lookups for that particular row.

There are two findings from the audit. The first one relates to location_dim table. The Insert/Update process, which is meant to populate this particular dimension table, was done using a surrogate key rather than a natural key. This means that each time the process is run, new surrogatekeyed records are added to the table and not the update existing ones. Consequently, running the process from a populated data warehouse would mean that there would be duplicate location dimensions records created, and thus the fact records generated during subsequent runs would refer to inconsistent location_code values, hence making province-level aggregation incorrect on the dashboard. The second finding relates to owner_dim table where deduplication process uses only the owner_title field rather than owner_title + kepemilikan combination.

F. Star Schema



The schema design for the warehouse uses a star schema approach wherein the fact_table refers to four dimension tables using surrogate foreign keys. Each individual dimension is loaded by a separate transformation process in Pentaho, which picks out the appropriate natural keys from the hospital_clean table and inserts the unique data into the MySQL database using Insert/Update, where the surrogate keys themselves (jenis_code, kelas_code, location_code, owner_code) are automatically assigned in MySQL as a BIGINT AUTO_INCREMENT column as opposed to being assigned in Pentaho. The transformation process fact_dim_ppt.ktr then obtains all four surrogate keys through a sequence of Database Lookups before a Select Values process assigns order to the final eight keys (id, jenis_code, location_code, owner_code, kelas_code, total_tempat_tidur, total_tenaga_kerja, total_layanan, nama) and outputs the data into the fact_table.

The schema allows for slicing and dicing the query in multiple dimensions based on the geographic location, type of facility, facility class, and ownership dimensions for the three quantitative measures of capacity. This is possible because all surrogate keys for each dimension are generated by the database, not by any application logic, and therefore, in theory, there should be no conflict of keys for fact_table.

The use of SCD Type 1 in managing dimensions is justified by the goals of this project because the attribute values of the dimension are reference data that change rarely and there is no need to trace historical changes in the attributes. If the future goals of the project involve the need to track changes in the attributes for example, the change in ownership or classification, the SCD pipeline may be extended from SCD1 to SCD2.

G. Orchestration



After beginning at a Start entry, the job executes hospital_clean, location_dim, jenis_dim, owner_dim,

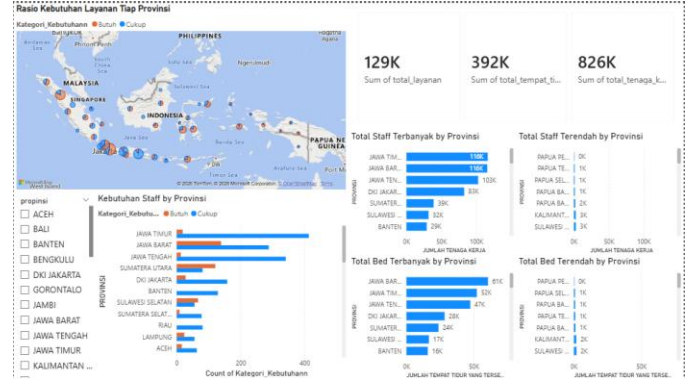
kelas_dim, and then fact_dim_ppt in sequential order, where each subsequent step depends on the success of the preceding transformation; the job ends at a Success entry only if all five transformations succeed without encountering any errors. The sequencing ensures that there is a crude mechanism for referential integrity by the order of buildsince all four dimensions will have been built before the fact_dim_ppt tries to look up any keys in them but it does not guarantee by itself that the dimensions are keyed properly or that the job can be re-executed safely if there already exists data in the warehouse. There is no external scheduler set up; the job is supposed to be started manually using either Pentaho Spoon or some other scheduling mechanism like cron.

H. Consumption Layer/Serving Layer(Power BI)

The Power BI Desktop model is directly connected to the ppt_olap schema using the MySQL connector, importing all five tablesfact_table, together with its four dimension tables. The star schema relationships between these tables are established via the surrogatekeys that are created when loading data into the data warehouse. Seven DAX measures are defined to meet the needs of the dashboard analysis. Among them, there are three basic aggregations: Total Beds (sum of total_tempat_tidur), Total Workforce (sum of total_tenaga_kerja), and Total Services (sum of total_layanan). Another measure provides the count of hospitals for the current filter context. Based on these measures, two other measures are calculatedaverage number of beds per hospital that helps to see how different hospitals in terms of their class are in terms of efficiency, and workforce-to-bed ratio that is the main equity indicator from the side of the supply.A seventh measure called Kategori Kebutuhan defines the binary classification of provincesSufficient or Needbased on the same workforceto-bed ratio

The dashboards are arranged on eleven report pages. The first page offers a national view with KPI cards for the three aggregated metrics. Following pages show geographic distribution map based on colorcoded Kategori Kebutuhan, provincelevel ranking of workforce and beds capacity, types of facilities and hospitals, types of ownership, financial state of BLUs/BLUDs, comparison of the best and worst provinces and need/sufficient classification summary. The eleventh page is a hospital details matrix providing drilling capability to the specific records. Interactive province slicers are available on all pages making crossdimensional slicing possible without changing any DAX formulas. This is how the design was done according to the recommendations of Nurhakim and Voutama [12]. It is said that the following visual elements are most efficient in communicating healthcare warehoused information: interactive slicers, KPI cards and ranked bar charts.

IV. RESULT AND DISCUSSION



At the national level, healthcare availability in Indonesia shows an acceptable workforceto-bed ratio of 2.1, backed up by a total of 826K healthcare workers, 392K beds, and 129K services. Nonetheless, these numbers mask a severe distributional disparity between provinces and districts; it becomes evident that there is an extremely strong concentration of resources in the island of Java. Higher populations and higher service volumes in the region are represented by bigger pie charts, mainly showing a "Sufficient" (blue) status but some "Need" (orange) status as well. Such a concentration can be seen from the fact that five out of seven provinces having the largest number of healthcare workers belong to Javanamely East Java, Central Java, West Java, Banten, and also DKI Jakarta. The same goes for the inpatient capacity, where West Java dominates with up to 61K beds.

On the other hand, signs of inequality in the delivery of healthcare services in Indonesia are evident in regions like Maluku, East Kalimantan, and especially Papua, whose smaller-sized pie charts illustrate lower service deliveries. The deficiency of resources for healthcare provision in Papua occurs in two ways, namely, in terms of medical staff and beds. It is important to mention that Highland Papua, a recently created autonomous province, is experiencing a critical lack of health workforce almost at the point of zero (0K). Such contrasting picture of Java and eastern parts of Indonesia illustrates the disparity in healthcare distribution in the country.

The star schema arrangement of the datawarehouse enables the reconsideration of these equity observations at more granular levels without having to conduct an entirely new statistical analysis. For example, through the filtering of the dashboard according to the hospital class, one can determine if the staff concentration observed in Java is due to the Class A national referral hospitals that tend to be located in big cities because of legal requirements or whether it also holds true for Class C and Class D district-level facilities. The use of ownership filters will then allow for a head-to-head comparison of the capacity between the government-owned and private facilities in the same province, an issue directly relevant to JKN partnership considerations [8].

It is important to acknowledge that the two dataquality findings have a bearing on the reliability of these results. The unkeyed upsert in `location_dim`, does not affect the accuracy of the current analysis because the pipeline was executed only once against a clean starting state; however, any future rerun of the pipeline without correcting the naturalkey configuration would introduce duplicate location surrogate rows and cause provincelevel aggregations to be incorrect. The grain mismatch in `owner_dim`, may have partially reduced the granularity of ownershipdimension analysis in the dashboard, as distinct kepemilikan subtypes could have been collapsed to the same owner surrogate key. The evidence of mitigation observed in the live Power BI connection suggests this was at least partially corrected before the dashboard was built, but a full endtoend rerun with the corrected transformation is necessary to confirm that the ownershiplevel results are fully accurate.

V. EVALUATION AND REFLECTION

A. Tools and Technologies

- Pentaho Data Integration (PDI)

The reason for choosing Pentaho as the ETL tool is its ability to offer a visual and interactive interface in terms of designing pipelines without having to write codes manually. The different transformations in this process, starting from CSV File Input, Replace in String, String Operations, Filter Rows, and Unique Rows, up to Table Output, can all be explicitly defined, thus allowing one to easily audit and document data flows. In addition, PDI offers the possibility of orchestrating transformations through the Job feature, where the five transformations of `location_dim`, `jenis_dim`, `owner_dim`, `kelas_dim`, and `fact_dim_ppt` can be executed sequentially.

- MySQL

The choice of MySQL as the storage platform was made because of its open source nature, light weight and ability to provide the relational structure that fits into the star schema. Two separate schemas were used: "hasil" as staging/bronze layer and "ppt_olap" as analytics/gold layer. This helps perform cleansing and warehouse loading tasks independently without fear of contaminating the initial data set. Another feature of MySQL that helped was the use of BIGINT AUTO_INCREMENT, which was used to create surrogate keys for all dimension tables automatically.

- Power BI Desktop

The choice of Power BI as the serving layer is based on the fact that Power BI allows the connection to MySQL, allows the creation of DAX measures in declarative form, and supports features of interactive visualization like slicers, KPI cards, maps, and ranked bar charts. The seven DAX measures have been created, which include Workforce-to-Bed Ratio and Kategori Kebutuhan and which will help classify the provinces dynamically without making changes in the warehouse data. The eleven report pages have been created.

- Star Schema (Dimensional Modeling)

The star schema approach was used based on the dimensional model proposed by Kimball [3], which consists of one fact_table that is linked to four dimension tables, namely `jenis_dim`, `kelas_dim`, `location_dim`, and `owner_dim`. The choice of using this schema is due to the structural and static nature of the data at the hospital, implying that there is no need for the database to be highly normalized like the snowflake schema.

B. Advantages, Trade off, and Limitation

- Pentaho Data Integration

Advantages: The pipeline is visual and easy to follow without reading code. Each step can be tested independently. The idempotency built into the ingestion stage through the Truncate Table option ensures that re-runs do not produce duplicate data in the staging table.

Trade-offs: Pentaho does not have built-in scheduling capability; the job must be triggered manually through Spoon or an OS-level scheduler such as cron. This means there is no automatic retry mechanism, alerting, or real-time pipeline observability of the kind offered by a dedicated workflow manager such as Apache Airflow.

Limitations: It was found that the Insert/Update step in `location_dim` was not configured with the correct natural key, meaning every re-run of the pipeline would insert new surrogate rows rather than updating existing ones. This is a configuration flaw that Pentaho itself does not detect and was only discovered through a manual post-load audit, highlighting the importance of independent validation beyond what the tool enforces by design.

- MySQL as the Warehouse Backend

Advantages: MySQL integrates easily with both Pentaho and Power BI. The separation of the two schemas hasil and ppt_olap provides a clear isolation between the staging and gold layers, allowing cleansing to be repeated without affecting the warehouse.

Trade-offs: MySQL is a transactional RDBMS, not a columnar database optimized for OLAP workloads. At a larger data scale, multi-dimensional aggregation queries would run significantly slower compared to solutions such as ClickHouse, BigQuery, or Amazon Redshift.

Limitations: No partitioning or specialized indexing strategy was applied to fact_table. At the scale of 3,147 records this presents no performance concern, but if the dataset were extended to cover all Indonesian healthcare

facilities longitudinally, this limitation would become significant and would require architectural reconsideration.

- Star Schema Design

Advantages: The star schema makes possible dynamic cross-dimensional queries, for instance, a comparison of bed capacity between various combinations of provinces, hospital class, and hospital ownership. The design also allows for easy extension, in case more analytical dimensions are needed in the future by simply adding new dimension tables.

Trade-Offs: The choice of SCD Type 1 for all the dimensions implies that there will be no historical record of any changes. In case there are any changes in the class or ownership of the hospital in future, there is no way to restore the historical status from this data warehouse.

Limitations: There was an issue identified with the dimensional data being too granular relative to the grain of the data. This was seen in the owner_dim table where de-duplication was done using the owner_title column only. Without the kepemilikan column included in the natural key, different sub-types of ownership could be represented by the same surrogate key.

- Power BI as the Serving Layer

Advantages: Interactive slicers applied globally across all eleven report pages allow users to explore the data without requiring any knowledge of SQL. DAX measures such as Kategori_Kebutuhan provide dynamic classification that is immediately visualized on the geographic map, making the equity findings accessible to policy audiences.

Trade-offs: Power BI Desktop is a local application and does not support real-time collaboration or web-based access without Power BI Service, which requires a paid license. Furthermore, the direct connection to MySQL means every data refresh requires an active database connection, which is not ideal for broader deployment scenarios.

Limitations: kelas_dim was not fully integrated into the Power BI model in the submitted version, and a duplicate Kategori_Kebutuhan column was present. This means the hospital class dimension could not be used as a filter in the dashboard, leaving one of the primary analytical dimensions unexploited in the current state of the serving layer.

- The Decision to Use ETL over ELT

Advantages: Choosing ETL ensured that data entering the warehouse was already clean and standardized. The eight defective records were identified and excluded before reaching the dimensional model, preserving the quality of data in the gold layer and preventing bad data from propagating into the dashboard.

Trade-offs: The ETL approach is less flexible than ELT if re-transformation with different rules is later required, because raw data in the staging layer is not retained in a form that is easy to re-transform without re-running the entire ingestion step from the beginning.

Limitations: The current pipeline is not incremental every run processes the entire dataset from scratch. For a dataset that is updated periodically, an incremental load strategy or Change Data Capture (CDC) mechanism would need to be implemented to keep the pipeline efficient as data volume grows over time.

VI. CONCLUSION

In this paper, a full data engineering pipeline was created for the national registry of hospitals in Indonesia: ETL with Pentaho Data Integration, star schema warehouse on MySQL, and visualization via Power BI. Moreover, this paper included the evidence-based validation of the obtained warehouse instead of mere assumptions about its quality. It was demonstrated that cleansing eliminated exactly eight rows due to the errors in the source data. As for the referential integrity audit, there were two identified problems the unkeyed location_dim upsert and the miskeyed owner_dim deduplication, as well as the evidence of mitigation of the second problem in the system used to feed the dashboard. In this regard, it was proved that there is a great discrepancy between healthcare infrastructure of Java province and the eastern ones, as Papua provinces suffer from a severe shortage in both personnel and beds.

There are at least four ways in which this system could be further developed. First, implement correctly keyed upserts on all dimensions, as well as truncate or upsert on hospital_clean and fact_table and then run the whole job to verify that the reference integrity problems have been fixed, not just in the Power BI connection. Second, add dynamic utilization metrics like Bed Occupancy Rate, Average Length of Stay, Turn Over Interval, and Bed Turn Over [9] to complement the capacity metrics already being used to differentiate between the provinces that lack capacity and those that lack utilization. Third, bring back kelas_dim into the Power BI model and fix the problem with duplicated Kategori_Kebutuhan columns, making sure that both the hospital class and need classification metrics are being utilized in the dashboard. Fourth, substitute

the manual execution of the Pentaho job with a fully orchestrated process managed via Airflow, including CSV ingestion step.

VII. REFERENCES

[1] M. Kleppmann, Designing DataIntensive Applications. Sebastopol, CA: O'Reilly Media, 2017.

[2] J. Reis and M. Housley, Fundamentals of Data Engineering. Sebastopol, CA: O'Reilly Media, 2022.

[3] R. Kimball and M. Ross, The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling, 3rd ed. Indianapolis, IN: Wiley, 2013.

[4] W. H. Inmon, Building the Data Warehouse, 4th ed. Indianapolis, IN: Wiley, 2005.

[5] T. White, Hadoop: The Definitive Guide, 4th ed. Sebastopol, CA: O'Reilly Media, 2015.

[6] Kementerian Kesehatan Republik Indonesia, "Data Fasilitas Pelayanan Kesehatan," Pusdatin Kemkes, Jakarta, 2023. [Online]. Available: <https://sirs.kemkes.go.id>

[7] Pentaho Data Integration Community, "Pentaho Data Integration (PDI) Documentation," Hitachi Vantara, 2023. [Online]. Available: <https://help.pentaho.com>

[8] Misnaniarti, B. Hidayat, Pujiyanto, M. Nadjib, H. Thabrany, P. Junadi, Besral, B. Purwoko, Trihono, and V. Yulaswati, "Ketersediaan Fasilitas dan Tenaga Kesehatan dalam Mendukung Cakupan Semester Jaminan Kesehatan Nasional," Jurnal Penelitian dan Pengembangan Pelayanan Kesehatan, vol. 1, no. 1, pp. 6-16, Aug. 2017.

[9] C. Mariana, A. Rois, A. Mukhsin, and M. Syarif, "Efisiensi Pendayagunaan Tempat Tidur Berdasarkan Indikator Pelayanan Kesehatan di Rawat Inap Rumah Sakit," Jurnal Menara Medika, vol. 8, no. 1, pp. 71-80, Sep. 2025.

[10] J. Purwanto and Renny, "Perancangan Data Warehouse Rumah Sakit Berbasis Online Analytical Processing (OLAP)," Jurnal Teknologi Informasi dan Ilmu Komputer (JTIIK), vol. 8, no. 5, pp. 1077-1088, Oct. 2021.

[11] I. P. W. Prasetya and I. N. H. Kurniawan, "Implementasi ETL (Extract, Transform, Load) pada Data warehouse Penjualan Menggunakan Tools Pentaho," TIERS Information Technology Journal, vol. 2, no. 1, pp. 39-47, Jun. 2021.

[12] I. Nurhakim and A. Voutama, "Analisis Efisiensi Pelayanan Kesehatan dengan Visualisasi Data Interaktif di Power BI," JITET Jurnal Informatika dan Teknik Elektro Terapan), vol. 13, no. 2, pp. 904-912, Apr. 2025.

VIII. ATTACHMENT

A. Similarity

166523-JASON ALEXANDER W.pdf

ORIGINALITY REPORT

1%SIMILARITY INDEX

1%INTERNET SOURCES

0%PUBLICATIONS

0%STUDENT PAPERS

PRIMARY SOURCES

1

edoc.sub.uni-hamburg.de

Internet Source

Exclude quotes

On

Exclude matches

< 1%

Exclude bibliography

On

B. AI

Jason Alexander W -

166523-JASON ALEXANDER W.pdf

Quick Submit

Quick Submit

LXC

Document Details

Submission ID

trncid:13603586046

Submission Date

Jun 29, 2026, 5:22 PM GMT+7

Download Date

Jun 29, 2026, 5:48 PM GMT+7

File Name

166523-JASON ALEXANDER W.pdf

File Size

374.4 KB

8 Pages

5,219 Words

28,974 Characters

turnitin

Page 1 of 10 - Cover Page

Submission ID: trncid:13603586046

turnitin

Page 2 of 10 - AI Writing Overview

Submission ID: trncid:13603586046

% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results, so it should not be used as the sole basis for adverse actions against a student). It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

